॥ श्रीगणेशाय नमः ॥

Vector Spaces and Matrices

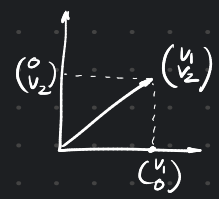
Algebra of vectors

Properties of vectors

① Length of vectors:

Say $v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$.

For a 2-D vector:



By the Pythagoras theorem,

$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$

(length of the vector)

$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$

② Scalar * vector:

$V = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \qquad \alpha V = \begin{pmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{pmatrix}$

③ vector + vector:

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}_{n\times 1} + \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ \vdots \\ u_n+v_n \end{bmatrix}_{n\times 1}$$

④ vector * vector → Inner product
→ Outer product

Inner product:

vector · vector (dot)

$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

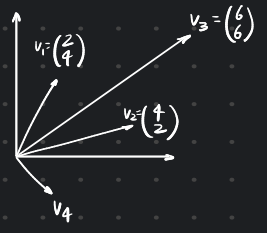
Dot product:

$u \cdot v = u_1v_1 + u_2v_2 + \cdots + u_nv_n$ (scalar)

$(n\times 1) \cdot (n\times 1)$

- Scalar product is distributive over vector addition: $\alpha(u+v) = \alpha u + \alpha v$
- Vector addition is commutative: $u+v = v+u$
- Dot product is commutative: $u \cdot v = v \cdot u$
- Dot product is distributive over vector addition: $u \cdot (v+w) = u \cdot v + u \cdot w$

Vector addition



$v_1 + v_2 = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}$ (v_1, v_2, v_3)

$v_1 - v_2 = \begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = v_4$

Vector addition and scalar multiplication

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = 1 \cdot \begin{bmatrix} a \\ b \end{bmatrix} + 1 \cdot \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} - \begin{bmatrix} c \\ d \end{bmatrix} = 1 \begin{bmatrix} a \\ b \end{bmatrix} + (-1) \begin{bmatrix} c \\ d \end{bmatrix}$$

Linear combination of vectors $s_1v_1 + s_2v_2 + s_3v_3 + \cdots + s_nv_n$

s → scalar
v → vector

$$s_1 \begin{bmatrix} a \\ b \end{bmatrix}_{2\times1} + s_2 \begin{bmatrix} c \\ d \end{bmatrix}_{2\times1} = \begin{bmatrix} s_1a + s_2c \\ s_1b + s_2d \end{bmatrix}_{2\times1}$$

for n vectors: $s_1v_1 + s_2v_2 + \cdots + s_nv_n = v_0$ $\quad s_i \in \mathbb{R}$ $(-\infty, \infty)$

s_iv_i → vector

Linear dependence: If we can represent v_0 as a linear combination of v_i's, with some s_i's being non-zero, then $(v_1, v_2, \ldots, v_n, v_o)$ are called linearly dependent (LD).

Q. $v_1 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$. Are v_1 and v_2 LD?

→ we cannot write either of them as a linear combination of the other.
they are linearly independent (LI).

Q. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \end{bmatrix}$.

→ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{5} * \begin{bmatrix} 5 \\ 10 \end{bmatrix}$ they are LD.

Q. $\underset{v_1}{\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}}, \underset{v_3}{\begin{bmatrix} 7 \\ 10 \\ 14 \end{bmatrix}}$.

→ $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + 2 * \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 14 \end{bmatrix}$ $\quad v_3 = v_1 + 2v_2 \Rightarrow$ LD.

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are a linearly independent (LI) set of vectors.

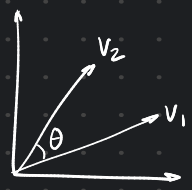
Q. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

→ $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Linear independence

A set of n vectors is called LI if none of the vectors can be written as a linear combination of the others.

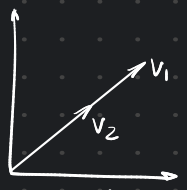
For $s_1v_1 + s_2v_2 + \dots + s_nv_n = 0$, we must have $s_1 = s_2 = \dots = s_n = 0$. ← LD if we get at least 1 $s_i \neq 0$.



Angle betⁿ the 2 vectors v_1 and v_2 is non-zero.

non-collinear

Independent



collinear vectors

Dependent

If every pair of vectors are non-collinear, (the set of vectors is linearly independent.
↳ or a linear combination of any 2 vectors is non-collinear with all other vectors in the set.

Atleast 1 pair is collinear → LD.
OR
lin. combⁿ of a pair is collinear to a 3rd vector.

For n vectors, if 2 of them are collinear, they are dependent. (or if LC of 2 is collinear to any of the others)

$s_1v_1 + s_2v_2 + s_3v_3 + \dots + s_nv_n = 0$

satisfied by (If v_i and v_j are collinear, $v_i = kv_j$

$s_1 = s_2 = \dots s_n = 0$ except $s_iv_i + s_jv_j = 0$

$\downarrow$

kv_j

$0\cdot v_1 + 0\cdot v_2 + \dots + (s_i)k\,v_i + s_jv_j + \dots + 0\cdot v_n = 0$

---

linear structure: follows 2 properties ↘ homogeneity: $f(a+b) = f(a)+f(b)$
↗ additivity: $f(sx) = sf(x)$

these are structures taken into consideration in linear algebra.

$f(x) = ax+b$. Is this a linear structure?

affine transformation

↳ additivity: $f(a+b) = f(a)+f(b)$
↳ homogeneity: $f(sv) = sf(v)$

→ $f(x+y) \stackrel{?}{=} f(x) + f(y)$

$f(x+y) = a(x+y)+b$

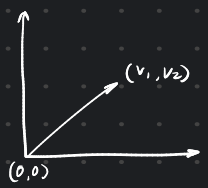
$f(x)+f(y) = ax+b+ay+b$
$= a(x+y)+b+b$
$= f(x+y)+b$

NOT a linear structure.

For our purposes: x is linear (strictly).

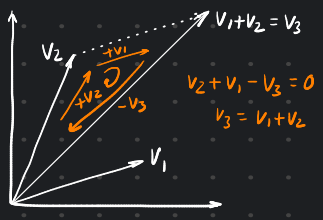
scaling: ax still linear (strictly linear)

translation: $ax+b$ ↳ not linear anymore



A vector is considered to start from $(0,0)$.

Geometric interpretation of vector addition:



$v_2 + v_1 - v_3 = 0$

$v_3 = v_1 + v_2$

Unit vector:

It is a vector with length = 1 in the original direction.

$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ $\quad \vec{v} \cdot \vec{v} = 1 = v_1^2 + v_2^2$

For any vector $v \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$,

the unit vector in the direction of v is $\frac{v}{\|v\|}$

Dot product:

$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ $\quad u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$

$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$

$\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$



Properties:

- $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$
- $|\vec{u} \cdot \vec{v}| \leq |\|\vec{u}\| \, \|\vec{v}\||$ or $-\|\vec{a}\| \cdot \|\vec{b}\| \leq \vec{a} \cdot \vec{b} \leq \|\vec{a}\| \cdot \|\vec{b}\|$
- $\cos\theta = \dfrac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \, \|\vec{v}\|}$
- $\|\vec{a} + \vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 + 2\vec{a} \cdot \vec{b} = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$
- $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{b}$

  $= \|\vec{a}\|^2 - \|\vec{b}\|^2$

$\begin{bmatrix}a\\b\end{bmatrix}, \begin{bmatrix}c\\d\end{bmatrix}, \begin{bmatrix}e\\f\end{bmatrix}$ are LI if none among them can be represented as a linear combination of the others.

$v_1 = \begin{bmatrix}1\\2\end{bmatrix}, v_2 = \begin{bmatrix}2\\4\end{bmatrix}$ $v_2 = 2v_1$ · LD set $\{v_1, v_2\}$

$v_1 = \begin{bmatrix}1\\2\end{bmatrix}$ $v_2 = \begin{bmatrix}2\\3\end{bmatrix}$ If $s_1v_1 + s_2v_2 = 0$ is not possible without $s_1 = s_2 = 0$, then $\{v_1, v_2\}$ is a LI set.

For $\{v_1, v_2\}$ to be LI, $s_1 = s_2 = 0$ is the only way for $s_1v_1 + s_2v_2 = 0$.

$v_1 = \begin{bmatrix}a\\b\end{bmatrix}$ $v_2 = \begin{bmatrix}c\\d\end{bmatrix}$

$$s_1v_1 + s_2v_2 = 0$$
$$s_1\begin{bmatrix}a\\b\end{bmatrix} + s_2\begin{bmatrix}c\\d\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

$$\left.\begin{aligned} s_1a + s_2c &= 0 \\ s_1b + s_2d &= 0 \end{aligned}\right\}$$ system of linear eqns.

$v_1 = \begin{bmatrix}0\\0\end{bmatrix}$ $v_2 = \begin{bmatrix}1\\2\end{bmatrix}$ $v_1 = 0v_2 \Rightarrow$ LD.

If the zero vector is a part of a set of vectors, the set is LD.

$\begin{bmatrix}1\\0\end{bmatrix}_{v_1}, \begin{bmatrix}0\\1\end{bmatrix}_{v_2}, \begin{bmatrix}2\\3\end{bmatrix}_{v_3}$ $\Rightarrow 2v_1 + 3v_2 + (-1)v_3 = 0$ or $v_3 = 2v_1 + 3v_2$

LD

Q. Are these LI/LD?

$\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}$

→ LI $\begin{bmatrix}1\\2\\3\end{bmatrix}$ we'd never get this component from the other 2.

In 2D space, there can be only 2 LI vectors.

meaning $\begin{bmatrix}2\\3\end{bmatrix}\begin{bmatrix}2\\5\end{bmatrix}$ can be LI,

but add any other vector,

$\begin{bmatrix}2\\3\end{bmatrix}\begin{bmatrix}2\\5\end{bmatrix}\begin{bmatrix}3\\6\end{bmatrix}$ are LD.

In n-dim space, n vectors can be LI at max.

Space:

$\mathbb{R}^n$ is a space with all possible n-dim vectors.

$\begin{bmatrix}1\\0\end{bmatrix}$ $\begin{bmatrix}0\\1\end{bmatrix}$ is a LI set. But adding any random vector to the set now would make it LD.

Any new vector like $\begin{bmatrix}15\\14\end{bmatrix}$ (in $\mathbb{R}^2$) can be represented as a linear combin^n of those 2.

$$\begin{bmatrix}15\\14\end{bmatrix} = 15\begin{bmatrix}1\\0\end{bmatrix} + 14\begin{bmatrix}0\\1\end{bmatrix}$$

Any vector in $\mathbb{R}^2$ can be represented as the linear combination of $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$.

Extension:

Any vector in $\mathbb{R}^2$ can be represented as a lin. comb^n of any 2 LI vectors in $\mathbb{R}^2$.
So we can fill $\mathbb{R}^2$ space by lin. combina^{n}s of any 2 LI vectors in $\mathbb{R}^2$.

We can fill the $\mathbb{R}^n$ space by taking linear combinations of any n LI vectors in $\mathbb{R}^n$.

NOTE: For 3 vectors, if 3 are coplanar (dim 2), the three are LD.

For $\mathbb{R}^n$, if all vectors lie in (n-1) dim subspace, they are LD.

Given 3 vectors:

		can they fill $\mathbb{R}^3$?
1) 2 vectors lie on a line	→ LD	NO
2) 2 vectors lie on a plane	→ could be LI	Potentially yes.
3) 3 vectors lie on a plane	→ LD	No
4) 3 vectors lie on a plane	→ LD	No for sure.

2 2D vectors lie on a line (1d)
3 3d vectors lie on a plane (2d)
4 4d vectors lie on a 3d surface
n nd vectors on an n-1 surface

} LD

Vector spaces

It is a set of vectors that is closed under vector addition and scalar multiplication.

$V = \{v_1, v_2, \cdots, v_n\}$

If $v_1, v_2 \in V$, then $v_1 + v_2 \in V$ and $sv_1 \in V$.

They must obey these properties:

a) closed under vector addition

b) closure over scalar multiplicat^n.

c) commutative over addition

$$v_1 + v_2 = v_2 + v_1$$

d) Associativity over +

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$$

e) Existence of zero (identity for addition)

$$v_1 + \underline{0} = v_1$$

f) Existence of inverse (for +):

$$v_1 + \underline{-v_1} = 0$$

g) Distributive over scalar multiplicat^n $(\cdot)$:

$$c(v_1 + v_2) = cv_1 + cv_2$$

h) Distributive

$$(c + d)v_1 = cv_1 + dv_1$$

i) Associativity over $(\cdot)$

$$(cd)v_1 = c(dv_1) = d(cv_1)$$

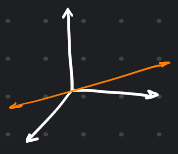
j) Existence of identity:

$$\underline{1}V = V$$

Examples of vector spaces:

1) $\left\{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right\}$ is a vector spaces.

2) The set of vectors on a line through the origin.



As the zero vector must be a part of the VS.

Span

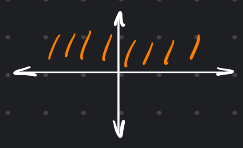
The span of a set of vectors is the space that can be achieved by taking the linear combinations of the vectors in the set.

Eg. $\mathbb{R}^2$ is the span of the set $\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$.

2. $\text{span}\left(\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}\right)$ = X-axis

3. $\text{span}\left(\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}\right\}\right)$ = X-axis.

Q. check if the foll^n are valid vector spaces.

① 

Everything on the +ve side of y-axis.

→ $(2, -3)$ Additive inverse $= (-2, 3) \notin V$. Not a valid VS.

② $(1, x, x^2, \cdots, x^n)$. Is this a VS for a set of polynomials?

Polynomial space: $s_1x^0 + s_2x^1 + s_3x^3 + \cdots + s_nx^n$

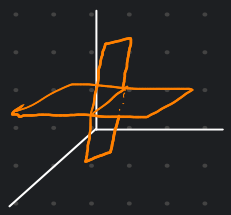
→ Yes. It is a valid VS.

③ Space spanned by intersection of 2 planes in $\mathbb{R}^3$.

④ Space spanned by union of 2 planes in $\mathbb{R}^3$.

} can be both, think of scenarios.

→ Intersec^n:



case I:

Intersec^n is line.

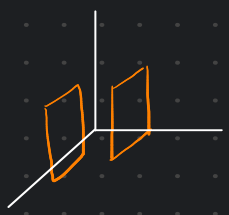
so if intersec^n passes through the origin, then yes.

case II:

2 planes are ||.

Intersection = ϕ.

Not a VS.



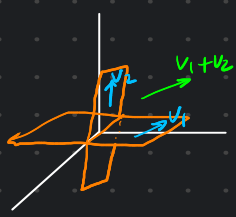
case III:

Intersection is plane.

If both the planes are the same, then yes.

If intersection is some other plane, NO.

Union of 2 planes:



Say 0 is a part of both VS.

So union will have 0.

But adding 2 vectors could go out of both planes so

NOT a valid VS.

only a valid VS if both are the same plane and dbr. passing through the origin.

Or if say one plane lies on an ∞ plane. Union will be that ∞ plane then if it has 0 then it is a valid VS.

Subspaces

A subspace is a subset of a vector space that is a vector space itself with the same operations of vector addition and scalar multiplication defined as on the original vector space.

As any subspace is a VS itself, it must have the zero vector.

To check subspace:
- check for closure over (+) and (·)
- check for zero vector
- check for additive inverse

Consider a vector space V:
- The zero vector is a subspace of V.
- V is also a subspace of V.

A line passing through the origin is a subspace of $\mathbb{R}^2$.
A plane in $\mathbb{R}^3$ passing through the origin is a subspace.

common for both VS and subspace:
These are defined over a field. E.g. $\mathbb{R}^2$ is defined over $\mathbb{R}$.

$\left\{ s \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; x, y \in \mathbb{R} \right\}$ (s: scalar multiplier) Is it is a vector space over $\mathbb{C}$? yes

$\left\{ s \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; x, y \in \mathbb{C} \right\}$ Is it a VS over $\mathbb{R}$? False.

Basis

A basis vector is one of the vectors in a set (basis) that defines a vector space.
A basis of a vector space V is a set of vectors that are linearly independent and span V.

E.g. $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^2$ so $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is a basis vector for $\mathbb{R}^2$.

Ily. $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is a basis vector of $\mathbb{R}^2$ for $\left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 9 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^2$.

$\left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \begin{pmatrix} 10 \\ 7 \end{pmatrix} \right\}$ is not a basis for $\mathbb{R}^2$.

$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$ is the standard basis for $\mathbb{R}^n$.

A basis for $\mathbb{R}^n$ will have exactly n vectors. If less, it won't span, if more then LD.

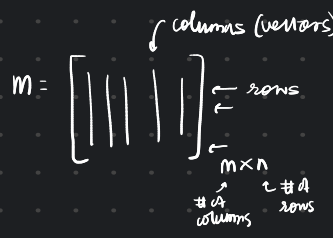
Dimension of a vector space

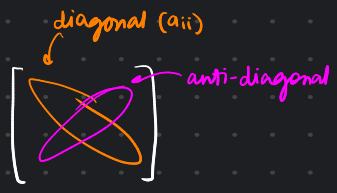
$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ vector space spanned is $\left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{R} \right\}$. $\dim_{VS} = 2$.

These are 3-dim vectors. But the vector space spanned by the set is 2 dimensional.

The dimension of a vector space V is the cardinality of a basis of V over its base field.

Matrix

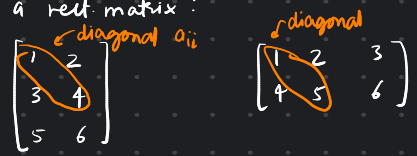




Square matrix : # of rows = # of columns

Rectangular matrix : # of rows ≠ # of columns

for a rect matrix :



Transpose of a matrix (m^T) :

a_{ij} = element in i^{th} row, j^{th} column.

for any element in m^T,

$$(m^T)_{ij} = m_{ji}$$

$$m = \begin{bmatrix} m_{11} & m_{12} \\ 1 & 2 \\ 3 & 4 \\ m_{21} & m_{22} \end{bmatrix}$$

$$m^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Identity matrix :

It is a sq. matrix with diagonal elements as 1.

$$m_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse of a matrix (m^{-1})

$$m\,m^{-1} = I$$

For a matrix to be invertible :

- it must be a sq. matrix
- $|m| \neq 0$

Diagonal matrix :

only diagonal elements can be non-zero.

$$\begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$$

$$\begin{cases} \overset{\text{atleast 1}}{m_{ij} \neq 0} & \text{if } i=j \\ m_{ij} = 0 & \text{otherwise} \end{cases}$$

Δar matrix :



Upper : all entries below diagonal are zero.

Lower : all entries above diagonal are zero.

(no restrictions on the diagonal itself: can be zero/non-zero)

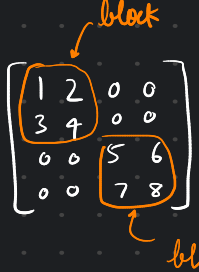
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$ → Upper Δar matrix

Block matrix :

sub-matrices/blocks

$$m_{4\times4} = \begin{bmatrix} [m_1]_{2\times2} & 0 \\ 0 & [m_2]_{2\times2} \end{bmatrix}$$

E.g. $$\begin{bmatrix} [1]_{1\times1} & 0 \\ 0 & [2]_{1\times1} \end{bmatrix}_{2\times2}$$



- has even # of rows and columns.

Q. Is it sufficient to say that if three 3-d vectors lie on a plane, they're linearly dependent?
Is it necessary for three 3-d vectors to lie on a plane for them to be LD?

→ $\begin{pmatrix}1\\2\\3\end{pmatrix}\begin{pmatrix}4\\5\\6\end{pmatrix}\begin{pmatrix}5\\7\\9\end{pmatrix}$ $v_1 \quad v_2 \quad v_3$

$v_3 = v_1 + v_2$ NOT on a plane but LD.

sufficient but not necessary.

Q. Given a set of LD vectors in R^3, can they fill the R^3 space?

→ If # of vectors ≤ 3 → NO

> 3 → yes, if atleast any 3 of them are linearly independent.

Properties of transpose

- $[A^T]_{ij} = A_{ji}$
- $(A+B)^T = A^T + B^T$
- $cA^T = (cA)^T$
- $(AB)^T = B^T A^T$
- $(A^{-1})^T = (A^T)^{-1}$

Proof: $(A^{-1})^T$

$$AA^{-1} = I$$
$$(AA^{-1})^T = I^T = I$$
$$(A^{-1})^T A^T = I$$
$$(A^{-1})^T A^T (A^T)^{-1} = I(A^T)^{-1}$$
$$(A^{-1})^T = (A^T)^{-1}$$

Symmetric matrix

$A^T = A$

- upper/lower Δar matrices cannot be symmetric.
- Diagonal matrices are always symmetric.

Skew-symmetric matrix

$A^T = -A$

$A_{ij} = -A_{ji}$, for non-diagonal entries.

For skew-symmetric matrices, the diagonal elements must be zero.

e.g. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Orthogonal matrix

$QQ^T = I$ or $Q^T = Q^{-1}$

Idempotent matrix

$A^2 = A$

e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Involuntary matrix

$A^2 = I \Rightarrow A = A^{-1}$

e.g. $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Nilpotent matrix

$A^K = 0$

e.g. zero matrix.

$A^2 = 0$ is sufficient to say $A^K = 0$

e.g. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ ← index = 2 as $A^2 = 0$

upto a certain K, A^K won't be zero. After that, it'll be zero. That K for which A is nilpotent is called the index.

$A^{K-1} \neq 0$ $A^{K+1} = 0$ $A^{K+2} = 0$

Matrix multiplication

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}_{2\times 2} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}_{2\times 1} = \begin{bmatrix} m_{11}a_{11} + a_{12}a_{21} \\ m_{21}a_{11} + m_{22}a_{21} \end{bmatrix}_{2\times 1}$$

For AB,
of columns in A = # of rows in B.

$A_{m\times n} \cdot B_{n\times p} = C_{m\times p}$

↳ $\begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix} a_{11} + \begin{bmatrix} m_{12} \\ m_{22} \end{bmatrix} a_{21}$

matrix (vector) = lin. combinⁿ of column vectors of matrix
coefficients of that lin comb. is decided by the values in the vector.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

m_1 m_2

$$\left[\begin{array}{c|c} m_{11} & m_{12} \\ m_{21} & m_{22} \end{array}\right] \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} | & | \\ C_1 & C_2 \\ | & | \end{bmatrix}$$

$C_1 = A_{11}m_1 + A_{21}m_2$

$C_2 = A_{12}m_1 + A_{22}m_2$

Q. $MA = B$. If the set of the columns of M are LI and A is a vector, then what can we say about B?

→ $\begin{bmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} | \\ A \\ | \end{bmatrix} = \begin{bmatrix} | \\ B \\ | \end{bmatrix}$

As LI, $m_1 s_1 + m_2 s_2 + m_3 s_3 = 0$ has only soln that all $s_i = 0$.
(s_1, s_2, s_3: scalars)

For $B = 0$ ⇒ some entries of $A \neq 0$.

Column space:
The space spanned by the columns of a matrix is the column space of the matrix.
Row space:
The space spanned by the rows of a matrix is the row space of the matrix.

$\begin{bmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{bmatrix}$ If column space = $\mathbb{R}^3$ ⇒ columns are LD.